Secuura Platform White Paper

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Document Authenticity on the Cardano Blockchain

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1. Executive Summary

Secuura is a document authenticity platform built on the Cardano blockchain. It enables organisations to certify the genuineness of documents and allows third parties to independently verify that a document has not been tampered with since its original certification. Secuura addresses the pervasive problem of document fraud, which costs the global economy an estimated $600 billion annually, by providing an immutable, blockchain-anchored proof of document authenticity.

What Secuura does: Secuura provides cryptographic proof that a document is genuine and unaltered. When an authorised issuer certifies a document, Secuura creates a unique cryptographic hash (fingerprint) of that document and permanently records it on the Cardano blockchain. Any subsequent attempt to verify the document compares a fresh hash against the blockchain record — any modification, however minor, will be detected.

What Secuura is not: Secuura is not a document storage solution, a digital signature tool, or an encryption service. It focuses exclusively on authenticity — proving a document is what it claims to be, issued by whom it claims to be issued by, and unmodified since certification.

Key differentiators:

- **Hash-based, document-less architecture** — Secuura never stores, transmits, or has access to original documents. Client-side hashing ensures documents never leave the user's device. This fundamentally eliminates the risk of document data breaches and simplifies compliance with data protection regulations.
- **Cardano blockchain immutability** — Certification records are anchored to the Cardano blockchain, providing a tamper-proof, publicly verifiable record that no single party can alter or delete. Cardano's proof-of-stake consensus mechanism ensures energy efficiency and low transaction costs.
- **Creator/Owner separation model** — The entity that issues a document (Creator) and the entity the document belongs to (Owner) are treated as distinct roles with different permissions. This reflects real-world document relationships — a university creates a degree, but the graduate owns and controls it.
- **Privacy-first compliance** — Full GDPR compliance including consent management, data subject request processing, right-to-erasure execution, and data portability. The hybrid on-chain/off-chain architecture ensures personal data can be managed per regulatory requirements while cryptographic proofs remain immutable.
- **Multi-level identity verification** — Five verification levels from basic email confirmation to government-issued identity checks, configurable per document type and per deployment jurisdiction.

Secuura targets organisations transitioning from legacy document management systems — universities issuing degrees, enterprises managing employment records, government agencies handling official documentation, and any sector where proving document authenticity to third parties is critical.

2. The Problem

2.1 The Scale of Document Fraud

Document fraud is a global challenge affecting every industry that relies on paper and digital records for critical decisions. Fraudulent academic credentials, forged employment records, counterfeit certificates, and tampered legal documents undermine trust in institutions and cause significant financial and reputational harm.

The fundamental problem is simple: How does a third party know that a document they have received is genuinely what it claims to be?

2.2 Current Solutions Are Inadequate

Approach	Limitation
Manual verification	Time-consuming, expensive, and error-prone. Requires contacting the issuing institution directly.
Digital signatures	Prove who signed, but not whether the underlying document is authentic or unmodified after signing. Signatures can be forged.
Centralised verification databases	Single points of failure. Can be hacked, corrupted, or discontinued. Trust depends entirely on the database operator.
QR codes and holograms	Can be copied, scanned from legitimate documents, and applied to fraudulent ones. Physical security features degrade.
Email or phone confirmation	Requires the issuer to maintain verification capacity. Not scalable. Susceptible to social engineering.

2.3 The Trust Gap

There is a fundamental asymmetry in document verification. The issuer of a document has high confidence in its authenticity at the time of issuance. But that confidence degrades through the document's lifecycle:

- Copies are made
- Documents change hands
- Organisations are restructured or closed
- Records are lost, corrupted, or altered
- Verification staff change
- Systems are decommissioned

By the time a verifier (an employer, a bank, a regulator) needs to confirm authenticity, the chain of trust between the original issuance and the current moment may be broken.

2.4 The Blockchain Opportunity

Blockchain technology offers a unique solution to the trust gap. By recording a cryptographic fingerprint of a document at the time of certification, and doing so on an immutable, publicly accessible ledger, it creates a permanent, tamper-proof reference point. Any modification to the document — however small — produces a different fingerprint, immediately revealing tampering.

3. The Secuura Solution

3.1 Principles

Secuura is built on five core principles:

1. **Originator-only certification** — Only the entity that created or issued a document can certify its authenticity. This preserves the trust chain from origin.
2. **Full provenance** — Every significant action in a document's lifecycle is recorded, creating a complete chain of custody from creation to the present day.
3. **Immutable records** — Certification records on the blockchain cannot be altered, deleted, or backdated by any party, including Secuura.
4. **Privacy by design** — The platform is designed to operate without ever seeing, storing, or transmitting original documents. Only cryptographic hashes are processed.
5. **Governance-driven** — Changes to platform rules require governance approval, and meta-rules (rules about rules) are immutable, preventing unilateral policy changes.

3.2 How It Works

```
DOCUMENT CERTIFICATION FLOW

1. ISSUER uploads document      → Hash computed client-side (BLAKE2b-256)
   to their local browser         Document NEVER leaves the device

2. CERTIFIER signs the hash     → Cryptographic proof of certification
   with their Cardano wallet      Non-repudiable witness signature

3. Hash is ANCHORED to          → Transaction recorded on Cardano
   Cardano blockchain              Permanent, public, immutable record

4. VERIFIER uploads document    → Fresh hash computed, compared to chain
   for verification                Match = authentic, Mismatch = tampered
```

3.3 The Hash-Based Architecture

The most important architectural decision in Secuura is that the platform operates on hashes, not documents. When a document is certified, the following occurs:

1. The document is provided to the client-side application (browser)
2. A cryptographic hash is computed locally using BLAKE2b-256
3. Only the hash is transmitted to the Secuura backend
4. The hash is anchored to the Cardano blockchain
5. The original document is discarded from memory

This means: - **No document data breaches** — There is no document data to breach - **No storage costs** — No document storage infrastructure required - **GDPR-friendly** — No personal data in documents is processed or stored - **Tamper-evident** — Any modification to the document changes the hash - **Universal** — Works with any file format (PDF, images, Office documents, etc.)

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4. Core Concepts

4.1 Creator vs Owner Model

Secuura distinguishes between the entity that creates a document and the entity that owns or controls it.

Role	Definition	Rights
Creator (Issuer)	Entity that creates and certifies the document	Certify authenticity, modify, revoke
Owner (Rights Holder)	Individual or entity the document belongs to	Control sharing, grant access, transfer ownership

In most real-world scenarios, the Creator and Owner are different entities. A university (Creator) issues a degree to a graduate (Owner). An employer (Creator) issues a reference letter about an employee (Owner). A government agency (Creator) issues a licence to a professional (Owner).

This separation ensures: - Only the Creator can certify authenticity (preserving the trust chain) - Only the Owner can control who sees and verifies the document (preserving privacy) - Rights can transfer without Creator approval (permissionless ownership transfer)

4.2 Certification Events

Every significant action in a document's lifecycle is recorded as a certification event. These events form an immutable provenance chain:

Event	Description	Blockchain Record
Issue	Document first created and certified	Hash + timestamp + certifier proof
Sign	Certifier signs the document hash	Wallet signature + transaction
Anchor	Hash recorded on Cardano blockchain	Transaction hash + block number
Verify	Third party verifies document	Verification result + timestamp
Share	Owner shares certification	Share record + recipient
Transfer	Ownership changes hands	Transfer record + new owner
Revoke	Certification cancelled	Revocation reason + timestamp

4.3 Chargeable Events

Certain actions are designated as chargeable events, forming the basis of Secuura's revenue model. Pricing is configurable per deployment:

Event	Default Price (ADA)	Default Price (USD)	Typical Payer
Certification issued	5.0	2.50	Issuer
Certification signed	2.0	1.00	Certifier
Verification (pass)	1.5	0.75	Verifier
Verification (fail)	0.5	0.25	Verifier
Document shared	0.5	0.25	Owner
Rights transferred	3.0	1.50	Transferor
Certification revoked	1.0	0.50	Certifier

5. Platform Architecture

5.1 Microservices Architecture

Secuura is built as a cloud-native microservices platform with 22 services (19 backend microservices + 3 frontend portals). All services are accessible exclusively through a unified Nginx gateway — only two ports are exposed externally: **8881** (Marketing Website) and **8882** (Nginx Gateway for all applications and APIs).

Service	Internal Port	Responsibility
API Gateway	8080	Central routing, rate limiting, security headers, request sanitisation
Originate	4000	Document certification, hashing, GDPR compliance, chargeable events
PRISM	4001	Decentralised Identity (DID) management via Atala PRISM
Wallet Connector	4002	CIP-30 wallet integration, CIP-8 message signing
Auth	4003	Authentication, JWT, MFA, SSO (Entra ID, Auth0)
Timestamping	4004	RFC-3161 compliant timestamps
Anchoring	4005	Cardano blockchain transaction submission
Security	4008	OWASP compliance, security audit logging
Analytics	4010	Platform analytics and usage metrics
KYC	4011	Microsoft Entra Verified ID integration for identity verification
Midnight	4012	Zero-knowledge proof selective disclosure via Cardano Midnight
M365 Integration	4013	Microsoft 365 / SharePoint integration
VC Issuer	4014	W3C Verifiable Credentials issuance
Staking	4015	Token staking and reward distribution
Referral	4016	Referral programme management
Governance	4017	DAO governance, proposal management, token-weighted voting
NFT Certificate	4018	NFT certificate minting on Cardano
Billing	4019	Stripe payment integration and chargeable events
Transfer	4020	Document ownership transfer and delegation workflows
Issuer Frontend	80	Issuer Portal (served at `/` via gateway)
Verifier Frontend	80	Verifier Portal (served at `/verify/` via gateway)
Admin Frontend	80	Admin Dashboard (served at `/admin/` via gateway)

> **Note:** All internal ports listed above are accessible only within the Docker network. External access is routed exclusively through the Nginx gateway on port 8882.

5.2 Hybrid On-Chain/Off-Chain Architecture

Secuura employs a hybrid storage model that balances blockchain immutability with privacy compliance:

On-Chain (Cardano Blockchain): - Document hash - Certification timestamp - Event hash - Issuer proof (wallet address) - Certificate ID

Off-Chain (Encrypted PostgreSQL Database): - User profiles - Full metadata - Access rules - Audit logs - Event details - Consent records

Why hybrid?

Concern	Solution
GDPR Right to Erasure	Personal data is off-chain and can be deleted
Transaction costs	Only hashes on-chain — minimal Cardano fees
Query performance	Fast reads from PostgreSQL; blockchain for verification
Privacy	Sensitive details encrypted in database
Immutability	Blockchain hashes prove integrity regardless of off-chain changes

5.3 Infrastructure

Component	Technology	Purpose
Database	PostgreSQL 15	Primary data store with SCRAM-SHA-256 authentication
Cache	Redis 7	Session storage, rate limiting, temporary state
Connection Pooling	PgBouncer	Database connection management for production
Containerisation	Docker / Docker Compose	Service orchestration and isolation
Reverse Proxy	Nginx	Frontend gateway (port 8882)
Blockchain	Cardano (Yaci DevKit for development)	Immutable certification records

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6. The Secure Verification Service (SVS)

6.1 Document Lifecycle

The SVS defines a structured lifecycle for document certifications:

```
DOCUMENT LIFECYCLE PHASES

Phase 1: CERTIFICATION
  Draft → Hashed → Pending Sign → Signed → Anchoring → Certified

Phase 2: VERIFICATION
  Certified → Verified (or Verification Failed)

Phase 3: RE-CERTIFICATION
  Verified → Pending Sign → Signed → Anchoring → Certified
  (Verifier becomes Certifier for next phase)

Terminal States:
  → Revoked (certification cancelled)
  → Expired (certification past expiry)
```

6.2 Phase Transitions

Each transition follows strict rules enforced by the platform:

From State	To State	Required Action
Draft	Hashed	Document hash computed
Hashed	Pending Sign	Certification request submitted
Pending Sign	Signed	Certifier signs with wallet
Signed	Anchoring	Transaction submitted to blockchain
Anchoring	Certified	Blockchain confirmation received
Certified	Verified	Verifier confirms authenticity
Certified	Verification Failed	Verifier's document doesn't match hash
Certified	Revoked	Certifier revokes certification
Any	Expired	Time-based expiration

6.3 Document Lineage Visualisation

Secuura provides an interactive graph-based visualisation of a document's complete history:

- **Nodes** represent certification events (issuance, signing, anchoring, verification)
- **Edges** show the progression through the document lifecycle
- **Colour coding** distinguishes chargeable events, blockchain confirmations, and user actions
- **Expandable/collapsible nodes** allow drill-down into event details

- **Date-based filtering** enables historical analysis
- **Evidence panels** display blockchain transaction links and verification data

The visualisation is powered by React Flow with Dagre layout, providing an intuitive directed-graph representation of the complete provenance chain.

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7. Blockchain Integration

7.1 Why Cardano

Secuura chose Cardano as its blockchain for several strategic reasons:

Factor	Cardano Advantage
Energy efficiency	Proof-of-Stake consensus uses fraction of energy vs Proof-of-Work
Low fees	Transaction fees typically under 0.20 ADA (~$0.10)
Native assets	NFT certificates without smart contract overhead
Formal verification	Haskell-based, mathematically verifiable
Metadata support	Native transaction metadata (Label 674) for document anchors
Governance	Built-in governance framework (Project Catalyst, CIP process)
Midnight sidechain	Zero-knowledge proof selective disclosure — operational at `/api/midnight/selective-disclosure/generate` and `/verify`

7.2 Anchoring Process

1. **Hash preparation** — Document hash and metadata packaged as transaction metadata
2. **Transaction construction** — Cardano transaction with metadata label 674
3. **Submission** — Transaction submitted to Cardano network via node API
4. **Confirmation** — Transaction confirmed in a block (typically 20-60 seconds)
5. **Recording** — Transaction hash and block number stored in database
6. **Verification** — Any party can verify the anchor on any Cardano explorer

7.3 Smart Contract Framework

Secuura's smart contracts are written in Aiken (Cardano's native smart contract language) and define:

- **Document types** — Classification of certifiable documents
- **Role permissions** — What each stakeholder role can do
- **Certification actions** — Valid certification and verification operations
- **Immutable rules** — Meta-rules that cannot be changed
- **Multi-signature governance** — Multi-party approval for rule changes

7.4 Network Support

Network	Purpose	Explorer
Mainnet	Production certifications	cardanoscan.io
Preprod	Pre-production testing	preprod.cardanoscan.io
Preview	Development testing	preview.cardanoscan.io
Devnet	Local development (Yaci DevKit)	Local explorer

8. Stakeholder Roles and Interactions

8.1 Role Definitions

Role	Description	Key Actions
Issuer	Organisation that initiates certification	Upload documents, create certification requests, assign certifiers
Certifier	Authorised signatory who certifies authenticity	Review requests, sign with wallet, revoke certifications
Subject	Entity the document is about	Receive notifications, view certifications, request revocation
Rights Holder	Entity that owns/controls the certification	Share certifications, transfer rights, grant access
Verifier	Third party checking authenticity	Upload document for verification, view results, receive certificates
System Admin	Platform administrator	Manage users, organisations, configuration, monitor system health

8.2 User Verification Levels

Secuura implements a tiered identity verification framework:

Level	Method	Typical Use Case
Basic	Email or SMS verification	Low-risk document access
Standard	Multi-factor authentication (TOTP)	Standard operations
Enhanced	Federated identity (Microsoft Entra ID, Auth0)	Enterprise users
High	Third-party KYC (Onfido, Jumio)	Financial documents
Government	100-point ID check, Gov.UK Verify	Legal documents, property titles

Verification requirements are configurable per document type, per organisation, and per deployment jurisdiction.

8.3 Platform User Roles

Role	Scope	Permissions
OWNER	Document-level	Full control of owned documents
ISSUER_ADMIN	Organisation-level	Prepare and submit certification requests
ISSUER_APPROVER	Organisation-level	Approve and sign certifications
VERIFIER	Platform-level	Verify certifications
ORG_ADMIN	Organisation-level	Manage organisation settings and members
SYSTEM_ADMIN	Platform-level	Full administrative access

9. Use Cases

9.1 Education — Academic Credentials

Scenario: A university certifies a graduate's degree. The graduate later applies for employment and the employer needs to verify the qualification.

Flow: 1. University registrar (Issuer) uploads the degree certificate 2. University dean (Certifier) signs the certification with their wallet 3. Hash is anchored to Cardano blockchain 4. Graduate (Rights Holder) receives certification and shares with employer 5. Employer (Verifier) uploads the same document and receives instant verification

Value: Eliminates weeks of manual verification contact between employer and university. Provides cryptographic proof that the document is genuine and unaltered.

9.2 Recruitment — Employment Verification

Scenario: A company issues an employment reference for a departing employee. A new employer needs to verify the reference is genuine.

Flow: 1. HR department (Issuer) creates reference document 2. Hiring manager (Certifier) signs certification 3. Employee (Rights Holder) shares certification link with new employer 4. New employer (Verifier) verifies instantly via Secuura portal

Value: Prevents reference fraud. Provides instant verification without contacting previous employer.

9.3 Professional Certifications

Scenario: A professional body certifies an accountant's qualification. Clients and regulators need ongoing verification.

Flow: 1. Professional body (Issuer) certifies qualification 2. Certification linked to professional's account (Rights Holder) 3. Clients and regulators (Verifiers) verify qualification status in real-time 4. If qualification is revoked, blockchain record updates accordingly

9.4 Property and Land Titles

Scenario: A government agency certifies property ownership. When property is sold, the title needs to transfer securely.

Flow: 1. Land registry (Issuer) certifies title 2. Property owner (Rights Holder) holds certification rights 3. On sale, rights transfer to buyer via permissionless transfer 4. Buyer's bank (Verifier) verifies title authenticity

9.5 Product Authenticity

Scenario: A manufacturer certifies the authenticity of luxury goods. Buyers need to verify products are genuine.

Flow: 1. Manufacturer (Issuer) creates product certificate 2. QR code linked to certification record 3. Buyer (Verifier) scans QR code to verify authenticity 4. Full provenance chain visible (manufacturer → distributor → retailer → buyer)

10. Privacy and Compliance

10.1 Privacy by Design

Secuura's architecture embodies privacy by design principles as mandated by GDPR Article 25:

Principle	Implementation
Data minimisation	Only cryptographic hashes processed; documents never stored
Purpose limitation	Data collected only for specified certification/verification purposes
Storage limitation	Automated retention policies with configurable expiration
Accuracy	Blockchain immutability ensures certification records are accurate
Integrity and confidentiality	Encryption at rest and in transit; access controls
Accountability	Comprehensive audit logging; consent records; deletion logs

10.2 GDPR Compliance

Secuura implements comprehensive GDPR controls, all fully operational via dedicated API endpoints:

Consent Management: - Purpose-specific consent records with version tracking - Consent expiration support - Consent withdrawal with timestamp - Consent validity checking

Data Subject Rights: - Right of access — Data export in JSON format - Right to rectification — DSR system with audit trail - Right to erasure — Cascading deletion across 9 data categories - Right to portability — Downloadable data export - Right to restriction — Processing restriction support - Right to object — Objection request handling

GDPR API Endpoints (all operational):

Endpoint	Method	Purpose
`/api/gdpr/consent`	POST	Record user consent
`/api/gdpr/consent/withdraw`	POST	Withdraw consent
`/api/gdpr/consent/check`	GET	Check valid consent
`/api/gdpr/consent/:userId`	GET	List all user consents
`/api/gdpr/dsr`	POST	Create Data Subject Request
`/api/gdpr/dsr/pending`	GET	List pending DSRs (admin)
`/api/gdpr/dsr/:dsrId`	PATCH	Update DSR status
`/api/gdpr/export/:userId`	GET	Export user data (Article 20)
`/api/gdpr/export/:userId/download`	GET	Download user data as file
`/api/gdpr/erasure/:userId`	POST	Execute right-to-be-forgotten
`/api/gdpr/retention`	GET	List retention policies
`/api/gdpr/retention/enforce`	POST	Trigger retention enforcement
`/api/gdpr/deletion-log`	GET	View deletion audit trail

Data Retention: - Automated retention enforcement (daily scheduler) - Configurable per data type - Deletion methods: hard delete, soft delete, anonymise - Deletion audit log

Database Tables: `consent_records`, `data_subject_requests`, `data_deletion_log`, `data_retention_policies` — all with UUID primary keys, foreign key relationships, and audit timestamps.

10.3 Blockchain and GDPR Compatibility

A key challenge for blockchain systems is GDPR's right to erasure. Secuura resolves this through its hybrid architecture:

- **On-chain data** consists exclusively of cryptographic hashes — these are not personal data under GDPR as they cannot be reversed to reveal the original document or individual
- **Off-chain data** (user profiles, metadata, access rules) is stored in PostgreSQL and can be fully deleted or anonymised on request
- When erasure is executed, the on-chain hash becomes an orphaned reference with no personal data linkage

11. Security Architecture

11.1 Authentication

Method	Use Case	Implementation
JWT tokens	API authentication	Access + refresh token pair; rotation support
Multi-factor authentication	Enhanced security	TOTP (Google Authenticator compatible); backup codes
Microsoft Entra ID	Enterprise SSO	OAuth 2.0 flow with tenant management
Auth0	Social/enterprise SSO	OpenID Connect integration
Cardano wallets	Blockchain operations	CIP-30 connection; CIP-8 message signing
API keys	Programmatic access	Scoped keys with rate limiting; hash-stored (not plaintext)

11.2 Application Security

Control	Implementation
Input sanitisation	XSS prevention, SQL injection detection, path traversal blocking, command injection prevention
Security headers	Helmet.js: CSP, HSTS (1 year), X-Content-Type-Options, X-Frame-Options, Referrer-Policy, Permissions-Policy
Rate limiting	Global: 100 req/min; Auth: 10 req/15 min; Verification: 200 req/min; Redis-backed with in-memory fallback
CORS	Whitelist-based origin control; credentials enabled
Brute-force protection	Exponential backoff (500ms–30s); Redis-backed; 5-minute lockout
Error handling	Standardised error classes; production-safe responses (no stack traces)
Parameterised queries	Prisma ORM prevents SQL injection

11.3 Infrastructure Security

Control	Implementation
Container isolation	Docker containers with network segmentation
Secret management	Environment variable-based; no hardcoded secrets
Database security	PostgreSQL with SCRAM-SHA-256; Redis with password authentication
Health monitoring	Service health endpoints; container restart policies
Graceful shutdown	Signal handling; connection draining
Logging	Winston structured logging; file transports in production; error tracking

12. Smart Contract Framework

12.1 Aiken Smart Contracts

Secuura's on-chain logic is implemented in Aiken, Cardano's purpose-built smart contract language:

Document Contract — Core certification logic: - Document status transitions (Draft → Certified → Verified) - Certifier authorisation verification - Hash integrity checking

Provenance Contract — Chain of custody: - Event recording - Provenance chain validation - Temporal ordering enforcement

Multi-Sig Contract — Governance: - Multi-party approval for configuration changes - Threshold signatures - Governance proposal execution

Platform Governance Contract — Meta-rules: - Platform-level rules enforcement - Immutable rule protection - Deployment-specific rule configuration

12.2 Immutable Rules

Certain rules are designated as immutable and cannot be changed even by governance: - Only the originator can certify authenticity - Every action creates a blockchain record - Hash algorithms cannot be downgraded - Audit trail cannot be disabled

13. Frontend Applications

13.1 Issuer Portal

The primary interface for organisations certifying documents: - **Dashboard** — Overview of documents, statistics, recent activity - **Document Management** — Upload, certify, monitor, share, revoke - **Document Lineage** — Interactive graph visualisation of document history - **Wallet Connection** — CIP-30 wallet integration for signing - **Analytics** — Certification volume, verification rates, usage trends - **Billing** — Chargeable events, pricing, payment management

13.2 Verifier Portal

Public-facing interface for document verification (accessible at `http://localhost:8882/verify/`): - **File Upload** — Drag-and-drop verification by document upload (POST `/api/verification/verify`) - **Document ID** — Verification by certification identifier (navigates to `/result/:documentId`) - **QR Code** — Camera-based QR code scanning for instant verification - **Results** — Displays verification checks (exists, hash valid, blockchain anchored, not revoked, signatures), issuer details, and blockchain proof (transaction hash, block height, confirmation date)

13.3 Admin Dashboard

Platform administration interface: - **User Management** — Create, disable, manage users and roles - **Organisation Management** — Organisation setup and configuration - **Document Types** — Configure certifiable document types - **Workflows** — Define certification workflows - **System Health** — Error monitoring, service health, GDPR DSR tracking - **Audit Logs** — Comprehensive activity logging - **API Keys** — Manage programmatic access - **Pricing Configuration** — Configure chargeable event pricing

13.4 Status Page

Public system status monitoring interface.

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14. Integrations and Extensibility

14.1 Current Integrations

Integration	Purpose	Protocol
Microsoft Entra ID	Enterprise SSO authentication	OAuth 2.0
Microsoft 365 / SharePoint	Document source integration	Microsoft Graph API
Auth0	Social and enterprise SSO	OpenID Connect
Stripe	Payment processing	Stripe API + Webhooks
Cardano wallets	Blockchain operations	CIP-30 / CIP-8
Atala PRISM	Decentralised identity	PRISM SDK
IPFS (Pinata)	NFT metadata storage	IPFS API
Onfido / Jumio	KYC identity verification	REST API

14.2 API-First Architecture

Secuura's entire functionality is accessible via RESTful APIs, enabling: - **Third-party integrations** — Any system can certify and verify documents - **White-labelling** — Custom frontends using Secuura's API - **Webhook notifications** — Real-time event notifications - **SDK support** — JavaScript/TypeScript SDK for rapid integration

14.3 Future Integrations

- **Midnight sidechain** — Zero-knowledge proofs for selective disclosure (operational — generate and verify endpoints available)
- **European Digital Identity Wallets** — eIDAS 2.0 compatibility
- **Government identity services** — Gov.UK Verify, Australian myGovID
- **Document management systems** — Additional DMS integrations
- **Mobile applications** — iOS and Android native apps

15. Chargeable Events and Revenue Model

15.1 Event-Based Pricing

Secuura's revenue model is based on chargeable events — specific actions within the certification and verification lifecycle that generate platform fees. Pricing is stored in a database and can be configured per deployment.

15.2 Pricing Tiers

Tier	Included Events	Additional Cost	Target
Basic	100/month	Per-event pricing	Small organisations
Standard	500/month	Discounted per-event	Medium organisations
Premium	2,000/month	Further discounted	Large organisations
Enterprise	Unlimited	Custom pricing	Enterprise clients

15.3 Revenue Streams

1. **Platform fees** — Per-event charges for certifications and verifications
2. **Subscription tiers** — Monthly/annual subscription with included event volumes
3. **Enterprise licensing** — Custom pricing for large-scale deployments
4. **White-label licensing** — Fees for white-labelled deployments
5. **API access** — Programmatic access tiers

16. Technology Stack

16.1 Backend

Technology	Purpose
Node.js / TypeScript	Service runtime and type safety
Express.js	HTTP framework
Prisma ORM	Database access and migrations
PostgreSQL 15	Primary data store
Redis 7	Caching, sessions, rate limiting
Winston	Structured logging
Docker / Docker Compose	Containerisation and orchestration

16.2 Frontend

Technology	Purpose
React 18	UI framework
TypeScript	Type safety
Vite	Build tooling
Material UI (MUI)	Component library
React Flow (@xyflow/react)	Document lineage graph visualisation
Dagre	Directed graph layout
React Router	Client-side routing

16.3 Blockchain

Technology	Purpose
Cardano	Layer 1 blockchain
Aiken	Smart contract language
Yaci DevKit	Local development network
BLAKE2b-256 / SHA-256	Cryptographic hashing
CIP-30 / CIP-8	Wallet integration standards

16.4 Security

Technology	Purpose
Helmet.js	Security headers
bcryptjs	Password hashing
jsonwebtoken	JWT authentication
express-rate-limit	Rate limiting
CORS	Cross-origin resource sharing

17. Development Roadmap

Phase 1 — Core Platform (Current)

- Document certification and verification
- Cardano blockchain anchoring
- JWT + MFA authentication
- Issuer, Verifier, and Admin frontends
- GDPR compliance (consent, DSR, erasure, portability)
- Error tracking and structured logging
- Chargeable events with configurable pricing
- Microsoft Entra ID integration
- CIP-30 wallet integration
- Document lineage visualisation

Phase 2 — Extended Capabilities

- Mobile applications (iOS, Android)
- Midnight sidechain integration (zero-knowledge proofs) — **implemented and operational**
- Ownership transfer workflows — **implemented** (simple, sale, inheritance, corporate types)
- Advanced KYC integration (government identity services)
- White-labelling for enterprise clients
- Self-hosted deployment option

Phase 3 — Governance and Scale

- On-chain governance (DAO) — **implemented** (auto-registration, token-weighted voting, min 50-char proposal descriptions)
- Smart contract rules engine
- Multi-chain support consideration
- Advanced analytics and reporting
- Token staking and rewards

18. Competitive Landscape

Solution Type	Examples	Secuura Differentiator
Digital signature platforms	DocuSign, Adobe Sign	Secuura focuses on authenticity, not signing workflows
Blockchain notarisation	Stampery, OriginStamp	Secuura provides full lifecycle management, not just timestamping
Document management	SharePoint, Box	Secuura is complementary — verifies documents stored elsewhere
Credential platforms	Accredible, Credly	Secuura is document-agnostic and supports any certification type
Identity solutions	uPort, Civic	Secuura focuses on documents, not identity alone

Secuura's unique position: A comprehensive document authenticity platform combining blockchain immutability, privacy-first architecture, regulatory compliance, and enterprise integration capabilities.

19. Conclusion

Secuura addresses a fundamental challenge in the digital economy: proving that documents are genuine and unaltered. By combining the immutability of the Cardano blockchain with a privacy-first, hash-based architecture, Secuura provides a solution that is simultaneously tamper-proof and privacy-compliant.

The platform's microservices architecture ensures scalability and maintainability. Its comprehensive compliance framework — spanning GDPR, ISO 27001, SOC 2, OWASP, and blockchain standards — provides the assurance required by regulated industries. And its flexible verification levels and configurable pricing make it adaptable to deployment in any jurisdiction and industry.

As document fraud continues to evolve in sophistication, the need for a trustworthy, independently verifiable source of truth becomes ever more critical. Secuura provides that source of truth — anchored in mathematics, recorded on blockchain, and designed for the real world.

20. Glossary

Term	Definition
Anchoring	Recording a document's cryptographic hash on the Cardano blockchain
BLAKE2b-256	Cryptographic hash function used as default in Secuura
Cardano	Third-generation proof-of-stake blockchain platform
Certification	The process of creating a verifiable record of document authenticity
Certification Event	A recorded action in a document's lifecycle (issue, sign, verify, etc.)
Chargeable Event	A certification event that incurs a platform fee
CIP-8	Cardano Improvement Proposal for message signing
CIP-30	Cardano Improvement Proposal for wallet dApp connection
Creator	Entity that creates and certifies a document (also called Issuer)
DID	Decentralised Identifier — a self-sovereign digital identity
DSR	Data Subject Request — a formal request under GDPR
Erasure	The right to have personal data deleted (GDPR Article 17)
Hash	A cryptographic fingerprint of a document's contents
Owner	Entity that owns and controls a certified document
Provenance	The complete history and chain of custody of a document
Rights Holder	Entity with ownership rights to a certification
SVS	Secure Verification Service — Secuura's core verification system
Verifiable Credential	A cryptographically secure, machine-verifiable digital credential
Verification	The process of confirming a document matches its blockchain-anchored hash

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Secuura Platform White Paper

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